

It is interesting to note that all the above battery chemistries have been known for at least the last 40 years, with the lead-acid ones being known for more than 150 years now. From an electronics circuits design standpoint, the protection mechanisms that we shall discuss apply to all types of secondary (or rechargeable) batteries. Some protections are required during the charging process, while others make sense only during the discharge process. Thus, some

protections are implemented as part of the charger, while others are implemented as part of the battery management system that oversees the charging and discharging process of the battery.

The protection mechanisms we shall discuss are listed under Table 1. It is important to note that these are listed for the complete battery and not for individual cells. Protection of each cell would be part of another article on battery management systems.

Before we get into the details, it would be prudent to discuss the typical charging stages for a battery. Generally, battery charging is divided into three distinct stages. These can then be subdivided into as many as seven different stages based on the state of charge (SoC) of the battery. The SoC is a figure of merit of a battery's charge capacity and is quite elusive to measure accurately. However, for a large part of the charge-discharge curve of the battery, the SoC can be estimated using the battery terminal voltage.

The three charging stages are

bulk, absorption, and float stages. Some battery chemistries are quite sensitive to impressed voltages and hence only expect very slight float voltage and usually require the charging circuit to disconnect once a certain terminal voltage is reached.

The most common charging algorithm used is the CC-CV method, which stands for constant current-constant voltage method. While some chemistries only require a CC method, the others get charged using only the CV method. Variations in the impressed voltage and the permissible charging current result in more charging stages.

Charging over-voltage protection

Charging over-voltage protection mechanism must ensure that during the charging process the battery is not subjected to voltages higher than the maximum recommended value.

While the battery is in the CV charging phase, over-voltage protection is a continuous control problem, where a complicated PID loop

is used to ensure that the voltage is fixed and does not exceed the voltage set point for that phase. Typically, this is achieved using a voltage regulator design, either using a discrete fixed or adjustable voltage regulator IC, or by using embedded code for a microcontroller which produces control signals to keep the voltage to the battery constant.

In the CC charging phase, over-voltage protection is a simpler limiting control problem. Here a comparator circuit in hardware or comparison operators in the microcontroller code can be used to ensure that the impressed voltage is lower than the maximum permissible voltage. Generally, both continuous control and limiting control functions are integrated into the charger to provide reliable control.

Charging over-current protection

This protection mechanism ensures that the current flowing into the battery is kept below a maximum permissible value. It is quite clear that one cannot push current into a

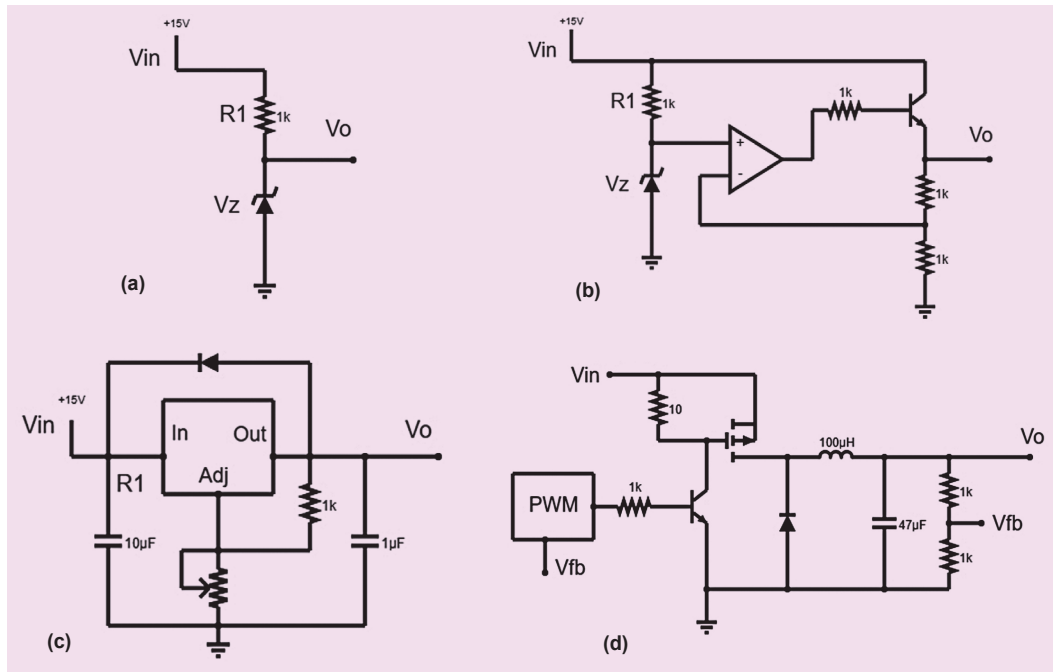


Fig. 1: (a) Zener voltage regulator, (b) op-amp and transistor regulator, (c) adjustable voltage regulator IC, (d) MOSFET switch based regulator